

ORIGINAL ARTICLE

Utilization and expenditure on medicines for the management of osteoporosis in Ireland: A repeated cross-sectional study

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Aim: Osteoporosis is a prevalent skeletal disease characterized by low bone mass and increased fracture risk. Management of osteoporosis typically involves antiresorptive and anabolic therapies, which are reimbursed in Ireland through various drug schemes. This study aims to summarize the utilization patterns associated with medicines used in the management of osteoporosis in Ireland.

Methods: This study is a repeated cross-sectional analysis of the overall utilization of and expenditure on medicines used in the management of osteoporosis in Ireland. A number of additional drug utilization metrics are investigated: the rate of osteoporosis medication prescribing per 1000 General Medical Services eligible population, adherence to bisphosphonates, denosumab and teriparatide, and the proportion of patients initiating denosumab as first-line treatment.

Results: There has been a significant change in the utilization of medicines used for the treatment of osteoporosis in Ireland over the last 13 years. The associated total annual expenditure was €28.7 million in 2011, increasing to €33.8 million in 2023. The most commonly used medicines have changed significantly over the period analysed; in 2011 the majority of patients were treated with bisphosphonates, whereas from 2019, the majority of patients were treated with denosumab. Poor patterns of treatment adherence were observed in this study.

Conclusion: The findings suggest evolving patterns in osteoporosis management in the Irish context. The study highlights the need for ongoing monitoring of medication use to ensure safe, effective and cost-effective care in the context of an aging population at increasing risk for osteoporosis.

KEYWORDS

bisphosphonates, denosumab, drug utilization, osteoporosis, teriparatide

1 | INTRODUCTION

Osteoporosis is a systemic skeletal disease that is characterized by low bone mass, micro-architectural deterioration of bone tissue

and compromised bone strength. This results in bone fragility and increased susceptibility to fracture.¹ Osteoporosis is the most common bone disease worldwide and is associated with high morbidity and mortality, as well as socio-economic costs.² Although the

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prevalence of osteoporosis currently is highest in North America and Europe, it will increase in developing countries as population life expectancy in these countries continues to increase.³

Medicines commonly used in the management, ie, treatment and prevention, of osteoporosis may reduce the rate of bone turnover (antiresorptive therapies) or stimulate bone formation (anabolic therapies). Antiresorptive therapies include bisphosphonates, denosumab, hormone replacement therapy and selective oestrogen receptor modulators (SERMs), while teriparatide has an anabolic or bone-forming action. A number of medicines for the management of osteoporosis are available for reimbursement in Ireland. In Ireland in 2005, over 87 000 patients received bisphosphonates, calcitriol, calcitonin, strontium, raloxifene or tibolone, with an associated expenditure of €22.12 million.⁴ In the same year, there were 606 prescription claims for teriparatide, with an associated expenditure of €1.99 million.⁴

Since 2005, the range of medicines reimbursed in Ireland for the management of osteoporosis has expanded to include the human monoclonal IgG2 antibody denosumab (2010) and the SERM bazedoxifene (2012).^{5,6} Internationally, osteoporosis treatment guidelines recommend using bisphosphonates first line in the management of osteoporosis.^{7–11} Some guidelines also support the use of denosumab as an alternative first-line treatment.^{10,11} Others recommend denosumab and alternative agents be reserved for patients for whom bisphosphonates are unsuitable or not tolerated.^{7–9}

The reimbursement price of certain medicines for the management of osteoporosis in Ireland has seen a reduction due to patent expiry and the introduction of generic medicinal products, and subsequent implementation of automatic substitution and reference pricing as legislated for in the Health (Pricing and Supply of Medical Goods) Act 2013.¹² In 2015 reference pricing was introduced for risedronate and alendronate, with a further reference price reduction for alendronate in 2022.^{13–15} Medicinal products containing a combination of alendronate and colecalciferol underwent a price reduction in 2016 as a result of reference pricing.¹⁶ Similarly, the addition of biosimilar medicines and hybrid medicinal products for teriparatide to the reimbursement list in 2019 triggered a price reduction in the originator brand in line with the provisions set out in the Framework Agreements between industry representative bodies and the health service.^{17,18}

The aim of this study was to determine the utilization and expenditure associated with medicines for the management of osteoporosis on the community drug schemes (CDS) and High Tech (HT) arrangement in Ireland. The demographic characteristics (age and gender) of people in receipt of these medicines are described and the duration of use of certain medicines used in the management of osteoporosis was determined.

2 | METHODS

2.1 | Data source

This retrospective repeated, cross-sectional study was performed using the Irish Health Service Executive-Primary Care Reimbursement

What is already known about this subject

- Osteoporosis is a common bone disease that is associated with high morbidity and mortality.
- A number of medicines are used in the management of osteoporosis, such as antiresorptive therapies and anabolic therapies.

What this study adds

- This study describes the changing landscape of osteoporosis management in Ireland, in terms of patient numbers and total associated expenditure, over a 13-period.
- This study highlights poor adherence to medicines used in the management of osteoporosis.

Services (HSE-PCRS) national pharmacy claims databases, which include the General Medical Services (GMS) scheme, the Long-Term Illness (LTI) scheme, the Drug Payment (DP) scheme and the HT arrangement. The HSE-PCRS is responsible for processing payments to healthcare practitioners such as general practitioners (GPs) and pharmacists for the provision of free or reduced-cost services to the public. The GMS scheme provides primary care services and hospital services free at the point of access. However, prescribed medications are subject to a co-payment. The current co-payment is €1.50 per item dispensed, capped at €15 per month per person or family. For patients over 70 years, the co-payment is €1.00 per item, capped at €10 per month per person or family. Patients are means tested to determine eligibility for the GMS scheme, with patients over 70 years having a higher income threshold than those under 70. Under the DP scheme, a person or family currently pays a maximum of €80 each month for approved prescribed drugs and medicines, and certain appliances. The LTI scheme provides free medicines, independent of income, to individuals with one of 16 chronic illnesses. The medicines available under the LTI scheme should be directly related to the treatment of the chronic illness, and a list of these can be found on the HSE website. The HT arrangement is administered by the PCRS, whereby HT medicines are purchased by the HSE and supplied through community pharmacies for which pharmacists are paid a patient care fee by the PCRS, with the cost of the medicines being paid by the PCRS directly to the manufacturer or wholesaler. Information on the number of HT prescriptions dispensed, the number of individual patients and the published ingredient cost is collected in the HT claims database.

2.2 | Data analyses

Data were included on patients from 1 January 2011 to 31 December 2023 who were dispensed medicines that are or were licensed for the

management of osteoporosis during the study timeframe: bisphosphonates (alendronic acid, ibandronic acid, risedronic acid), teriparatide, denosumab, SERMs (raloxifene, bazedoxifene), strontium ranelate and tibolone. Combination products, including a medicine for the management of osteoporosis with calcium or colecalciferol, were included in the analysis. Claims data relating to Xgeva[®], a branded version of denosumab licensed for use in malignancy, were excluded from the analysis. It should be noted that the pharmacy claims database does not contain information on prescribing indication and contains only basic patient information, such as age and sex.

Using the HSE-PCRS data, we identified the following factors. First, the utilization of osteoporosis medications, ie, the number of prescription claims, number of patients, breakdown of patient numbers by age category and sex for the most recent full-year dispensing. Second, the total state expenditure associated with osteoporosis medications exclusive of VAT and dispensing fees. Third, the rate of osteoporosis medication prescribing per 1000 GMS eligible population. The rate of prescribing was calculated for each year and presented as rate of prescribing per 1000 GMS eligible population. The number of patients using the DP scheme can vary over time depending on their reimbursement claim meeting the payment threshold, therefore the prescribing rate was only calculated for the GMS scheme, which has published eligibility data. Fourth, adherence to bisphosphonates, denosumab and teriparatide. These medicines were chosen for analysis as SERMs and tibolone may be used for non-osteoporosis indications. The proportion of days covered (PDC) method was used to estimate medication adherence by calculating the proportion of days in which a person has access to the medication over a given period of interest (eg, 28 days' supply of medication is assumed for a bisphosphonate claim).¹⁹ New users of denosumab, bisphosphonates and teriparatide in 2019 were included in this analysis, and their dispensed claims followed up for 2 years; patients did

not have a claim for the relevant medication in the year prior (2018). The PDC was calculated by dividing the total number of days covered with medication by the total number of days in the period of interest; 2 years from initiation of treatment. The PDC was measured as a continuous variable and reported in the text as a dichotomised variable according to whether PDC is at least 80% (adherent) or not (non-adherent). For denosumab and bisphosphonates, claims dispensed on the GMS scheme were included because the dispensing of these products on the DP scheme is dependent on patients meeting the payment threshold. For teriparatide, all claims dispensed on the HT arrangement were included. A sensitivity analysis was carried out using data from new users of denosumab, bisphosphonates or teriparatide in 2017 to account for the potential impact of the COVID-19 pandemic on medication initiation and continuation. Finally, the proportion of patients initiating denosumab as first-line treatment was calculated; for new-users of denosumab in 2017, the proportion of patients without a claim for a bisphosphonate (ATC Code M05BA or M05BB) in the 6 months and 12 months prior was estimated.

Analyses were performed using SAS 9.4 (SAS institute Inc.), R Studio and Microsoft Excel.

3 | RESULTS

3.1 | Utilization and expenditure on osteoporosis medications

There has been a significant change in the utilization patterns of medicines used for the treatment of osteoporosis in Ireland. Figure 1 illustrates the number of individual patients in receipt of medicines licensed for the management of osteoporosis on the CDS from 2011

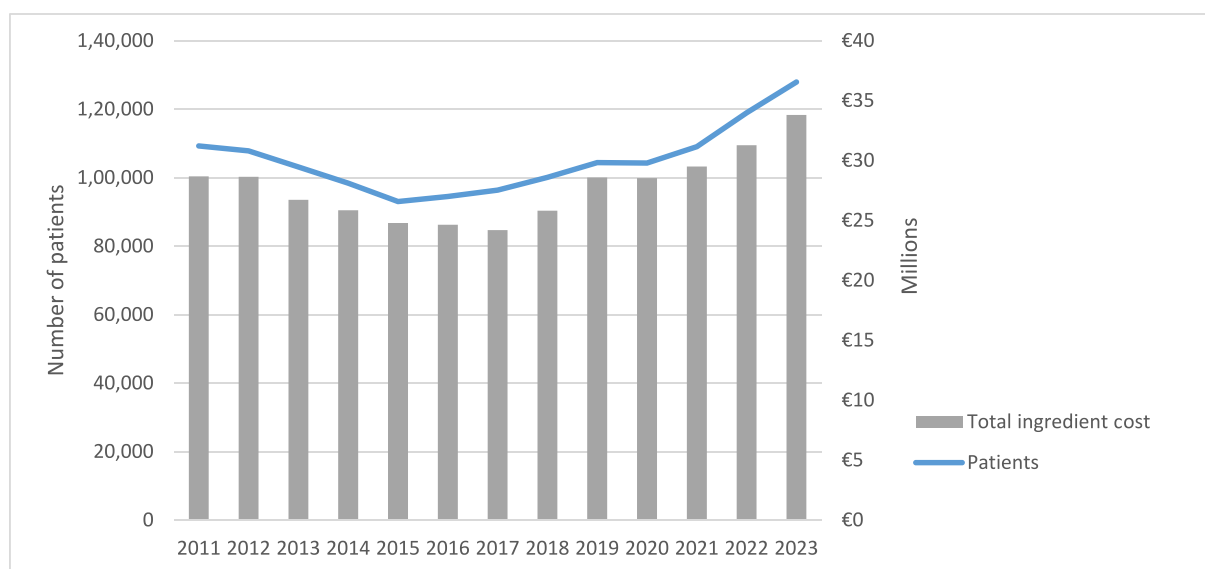


FIGURE 1 Total number of individual patients in receipt of medicines for the management of osteoporosis on the community drug schemes from 2011 to 2023 and associated total annual expenditure.

to 2023 and the associated total annual expenditure. The number of individual patients with claims for these medicines has increased slightly over the 13-year period, from 109 235 patients in 2011 to 127 887 in 2023. Total annual expenditure fell from €28.7 million in 2011 to just over €24.8 million in 2015, likely due to a decrease in eligible patient numbers over this time period, in combination with the introduction of reference pricing. Total annual expenditure increased from 2018 onwards, reaching €33.8 million in 2023.

Figure 2 outlines the utilization of the individual medicines from 2011 to 2023. The number of patients in receipt of denosumab has substantially increased, from just under 3000 patients in 2011 to over 75 000 patients in 2023. The number of patients in receipt of bisphosphonate treatment has declined over the same period, from 95 579 patients in 2011 to 52 755 patients in 2023. Of these, 41 000 received alendronate in 2011, reducing to 23 161 people in 2020 and increasing again to 32 053 in 2023. A similar reduction was seen in the number of people receiving medicinal products containing alendronate in combination with colecalciferol (vitamin D3): 19 694 patients in 2011 and 9043 in 2023. Similar patterns can also be seen for the other bisphosphonates. As strontium ranelate has been withdrawn from the Irish market, there is a significant reduction in the utilization of this medicine on the CDS. In 2011, there were 11 438 people in receipt of strontium ranelate on the CDS, fewer than 10 patients in 2022 and no patients in 2023. There has been an increase in the number of patients in receipt of teriparatide in Ireland; in 2011 there were just over 1600 patients in receipt of teriparatide and in 2023, just over 2600 patients.

In 2023, the majority of patients in receipt of medicines for the treatment of osteoporosis were female: 90.9% for denosumab, 80.8% for bisphosphonates, 98.9% for selective oestrogen receptor

modulators (SERMs), 99% for tibolone and 86.8% for teriparatide. In 2023, the majority of patients in receipt of denosumab, bisphosphonates, SERMs or teriparatide were aged over 70 years (Figure 3). The majority of patients in receipt of tibolone were aged 64 years and under.

Figure 4 shows the total annual expenditure on medicines for the management of osteoporosis on the CDS from 2011 to 2023. The total expenditure was €28.6 million in 2011 and €33.8 million in 2023. While the total annual expenditure has not changed significantly over this 13-year period, the distribution has changed significantly. For example, expenditure on denosumab accounted for €0.97 million in 2011 (less than 5% of the annual total), but had increased substantially to €26.4 million in 2023 (almost 80% of the annual total).

Figure 5 highlights the breakdown of expenditure on the individual bisphosphonates within the drug class. The total expenditure associated with bisphosphonates has decreased significantly over the study timeframe, as shown in Figure 4. In 2011, the total expenditure reached almost €18 million, in contrast to a total expenditure of just over €3 million in 2023. Alendronic acid, alendronic acid and colecalciferol combinations, and ibandronic acid comprised the majority of expenditure on bisphosphonates over the study timeframe.

3.2 | Prescribing rate of osteoporosis medication per 1000 GMS eligible patients

The prescribing rate per 1000 GMS eligible patients is shown in Table 1. There has been a large decrease in the prescribing rate of bisphosphonates from over 426.4 per 1000 GMS eligible in 2011 to

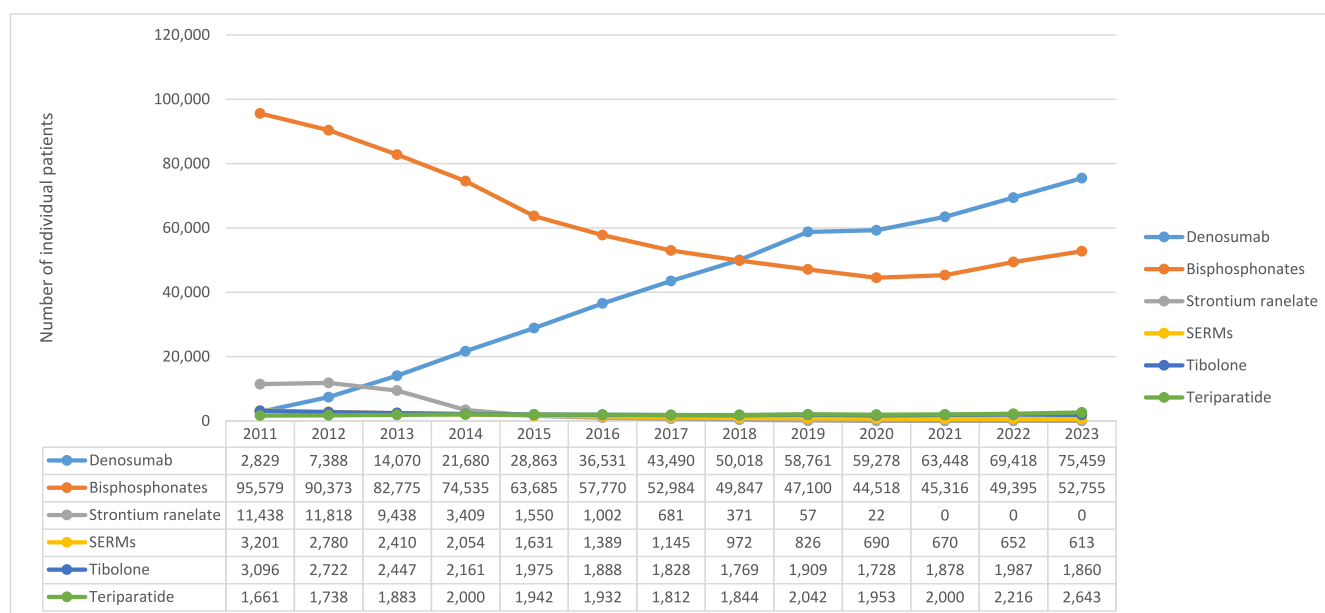


FIGURE 2 The number of individual patients in receipt of medicines for the management of osteoporosis on community drug schemes from 2011 to 2023.

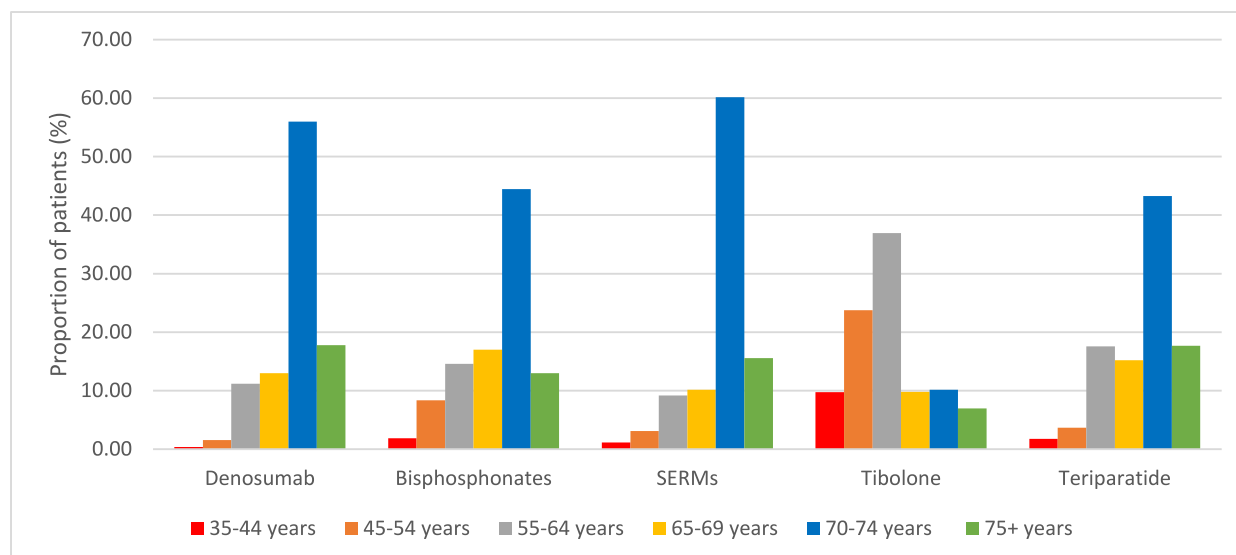


FIGURE 3 Age category breakdown of patients in receipt of medicines for the management of osteoporosis in 2023. SERMs, selective oestrogen receptor modulators.

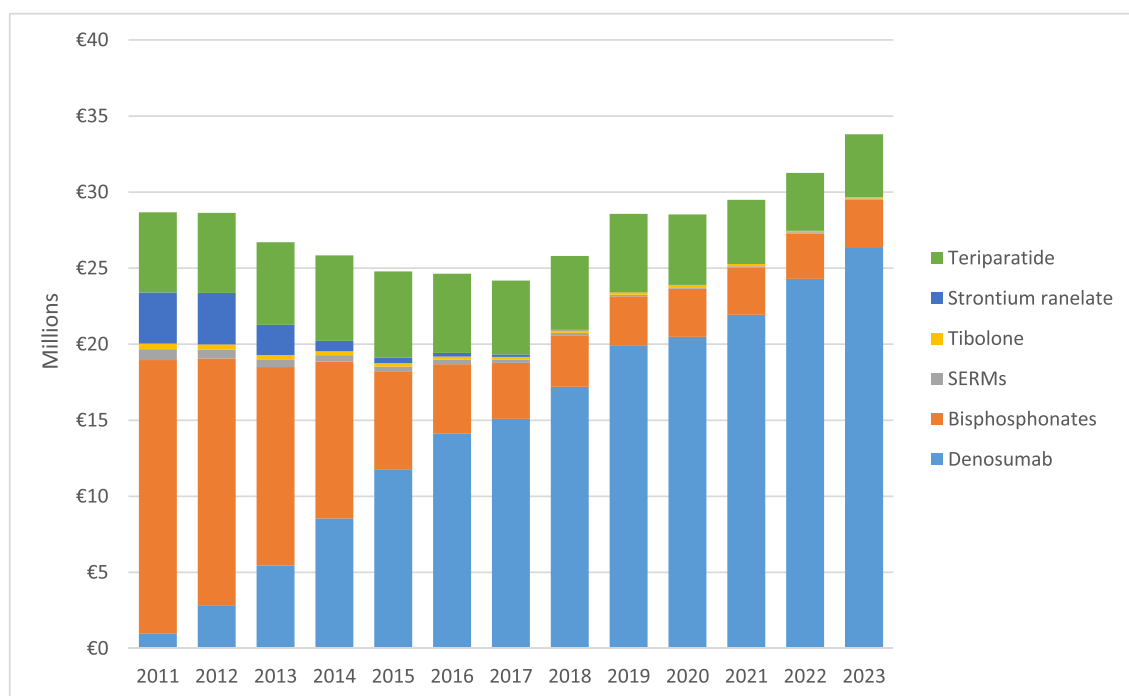


FIGURE 4 Total expenditure on medicines licensed for the management of osteoporosis on community drug schemes from 2011 to 2023. SERMs, selective oestrogen receptor modulators.

220.1 per 1000 GMS eligible in 2023. There has been an increase in the prescribing rate of denosumab over the study period, from 1.7 per 1000 GMS eligible in 2011 to 49.1 per 1000 GMS eligible in 2023. As the prescribing rate per 1000 GMS eligible is based on the number of claims per 1000 population with GMS eligibility, the apparently low prescribing rate of denosumab is reflective of the low number of claims dispensed per patient per year due to a dosing schedule of once every 6 months.

3.3 | Adherence to osteoporosis medication: bisphosphonates, denosumab and teriparatide

In 2019, there were 9952 new users of denosumab. The proportion of patients considered adherent to denosumab treatment 1-year post treatment initiation was high, at 73%. However, the proportion of patients adherent 2 years post treatment initiation was much lower, at 44%. Similar patterns of adherence were seen for bisphosphonates

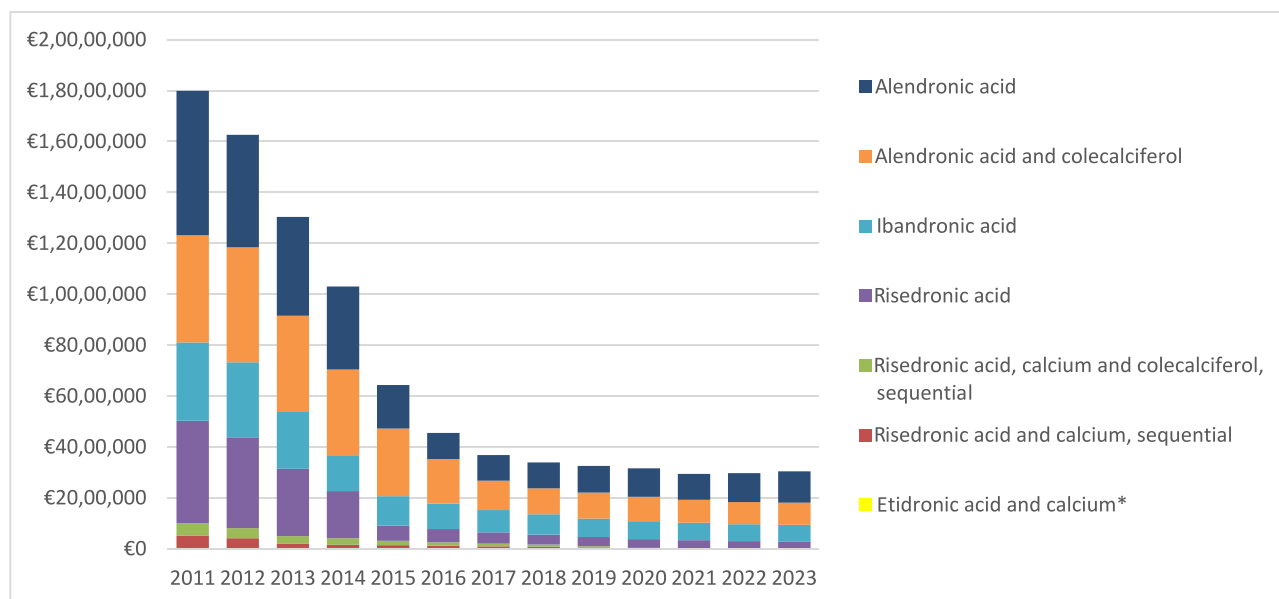


FIGURE 5 Total expenditure associated with bisphosphonates on community drug schemes from 2011 to 2023. *The marketing authorisation for etidronic acid and calcium (Didronel®) was withdrawn in Ireland in March 2010 and it was removed from the PCRS community drug schemes in March 2018.

and teriparatide. In 2019, there were 7588 new users of bisphosphonates; 1 year post treatment initiation the proportion of patients considered adherent was 60%. The proportion of patients considered adherent to bisphosphonates 2 years post treatment initiation was 49%. In 2019, there were 944 new users of teriparatide; the proportion of patients considered adherent 1 year post treatment initiation was 65%, decreasing to 43% at 2 years post treatment initiation. The results of this analysis are shown in Table 2.

Due to potential disruptions to medicines utilization during the COVID-19 pandemic, we repeated this adherence analysis using data from new users of osteoporosis medicines in 2017. In 2017, there were 8951 new users of denosumab; the proportion of patients considered adherent to denosumab treatment 1 year post treatment initiation was high at 75%, with the proportion of patients' adherent 2 years post treatment initiation at 47%. In 2017, there were 7607 new-users of bisphosphonates; 1 year post treatment initiation the proportion of patients considered adherent was 56%. The proportion of patients considered adherent to bisphosphonates 2 years post treatment initiation was 42%. In 2017, there were 783 new users of teriparatide; the proportion of patients considered adherent 1 year post treatment initiation was 67%, decreasing to 41% at 2 years post treatment initiation. The results of this analysis are shown in Table 3.

3.4 | First-line use of denosumab for the treatment of osteoporosis

In 2017, there were 8951 new users of denosumab. Of these, 2401 (26.8%) patients had a claim for a bisphosphonate in the 6 months prior to denosumab initiation and 2773 (31%) patients had a claim for

a bisphosphonate in the 12 months prior to denosumab initiation. This suggests that up to 70% of patients are being treated with denosumab first line. However, it should be noted that this analysis considers GMS claims only.

4 | DISCUSSION

To our knowledge, this is the first Irish study of the utilization and expenditure on medicines for the management of osteoporosis using national-level dispensed claims data between 2011 and 2023. There has been a significant change in the utilization of medicines for the management of osteoporosis over the timeframe, with an increase in the utilization and expenditure associated with the monoclonal IgG2 antibody, denosumab.

While there has been an overall decrease in the number of patients in receipt of bisphosphonates, there has been an increase in patient numbers since 2021; this is most likely due to a reduction in the payment threshold for the DP scheme. In November 2020, the DP scheme payment threshold was reduced from €124 to €114 per month. The threshold was further reduced to €100 and subsequently €80 in January 2022 and March 2022, respectively. This results in more patients meeting the payment threshold and thus being captured in the dispensed claims database. The lack of increase in the prescribing rate of bisphosphonates on the GMS scheme also suggests that the increase in patient numbers may be due to an increased number of individuals meeting the threshold for reimbursement support under the DP scheme.

The patient demographics seen in this study are similar to those observed in other Irish studies of osteoporosis treatments.^{4,20} A recent document by the International Osteoporosis Foundation

TABLE 1 Prescribing rate per 1000 General Medical Services eligible patients from 2011 to 2023.

	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
Denosumab	1.71	4.67	9.28	14.61	19.58	25.02	28.78	34.57	39.15	38.80	42.34	46.07	49.08
Bisphosphonates	426.43	381.98	346.79	317.74	286.51	265.23	251.08	241.43	229.83	221.46	223.22	218.20	220.07
Strontium ranelate	35.78	35.66	25.30	9.21	4.97	3.43	2.46	0.59	0.10	0.04	0.02	0.01	0.00
SERMs	12.88	10.45	9.34	8.23	6.91	6.07	5.33	4.76	4.17	3.54	3.32	3.02	2.68
Tibolone	8.27	7.33	6.95	6.55	6.22	6.34	6.23	6.25	6.31	6.28	6.43	6.22	5.63

estimated that 209 000 people in Ireland have osteoporosis, 77.5% of whom are female.²¹ This document also highlights that the Irish population aged 50 years or more is projected to increase by 38% between 2019 and 2034, which is higher than the EU average of 11.4%. The increases in the numbers of men and women aged 75 years or more are even more marked: 78.9% for men and 69.0% for women.²¹ As such, the number and burden of fragility fractures are likely to increase, and utilization of agents for the management of osteoporosis is likely to continue to increase.²¹ The Irish Central Statistics Office highlights that Ireland has had an aging population over the last 10 years, with the proportion of people aged 45 or over increasing from 35.4% in 2013 to 40.6% in 2023.²²

In 2011, the average annual cost per patient on bisphosphonates was approximately €188, which is in contrast to 2023, which had an annual cost per patient of approximately €60. This is likely driven by patent expiries, the introduction of generic medicinal products and reference pricing, which was introduced for risedronate in January 2015 and alendronate in May 2015, and for medicinal products containing a combination of alendronate and colecalciferol in August 2016.^{13,14,16} The reference price for alendronate was revised in February 2022.¹⁵ In Ireland, The Health (Pricing and Supply of Medical Goods) Act 2013 provides for the introduction of reference pricing, which involves the setting of a common reimbursement price, or “reference price”, for a group of medicines that are deemed interchangeable by the Health Products Regulatory Authority (HPRA).¹² In addition, under the Health Act 2013, medicines on the interchangeable list may be substituted for each other to create financial efficiencies for the health service. The publication of an interchangeable list by the HPRA facilitates automatic substitution by allowing pharmacists to safely dispense a lower-cost alternative to a medicine where a specific brand has been prescribed.

The savings generated from the introduction of reference pricing for bisphosphonates appear to have been negated by the increased utilization and expenditure associated with the uptake of denosumab, as demonstrated in Figure 5. The number of patients in receipt of denosumab has increased from 2800 patients in 2011 to over 75 000 patients in 2023, reaching a total expenditure of almost €26.4 million in 2023. In 2023, almost 60% of patients in receipt of medicines for the management of osteoporosis were treated with denosumab. The annual cost per patient on denosumab, based on ingredient cost alone and a twice-yearly dosing schedule, is approximately €400 per year. A study from the United States has reported the utilization of denosumab as having increased since its launch in 2010, accounting for approximately 20% of almost 950 000 patients treated for osteoporosis in March 2020.²³ In Australia, the data show a greater uptake in its utilization, with denosumab comprising 76.1% of patients dispensed any osteoporosis medicine in 2018 on the Pharmaceutical Benefits Scheme.²⁴

Financial efficiencies were also created through a best-value medicine (BVM) initiative. Where a biosimilar medicine and/or hybrid medicinal product become available for a biological medicine, such as teriparatide, the Medicines Management Programme may evaluate the biological medicine in question and issue a recommendation as to the BVM.¹⁸ Since 1 March 2023, it has been HSE policy that all adult

TABLE 2 Proportion of days covered for patients initiating treatment in 2019.

	Number of patients initiated in 2019		Mean PDC	Standard deviation	Proportion adherent (PDC ≥ 0.8)
Bisphosphonates	7588	Year 1	0.77	0.32	0.60
		Year 2	0.58	0.41	0.49
Denosumab	9952	Year 1	0.93	0.32	0.73
		Year 2	0.56	0.45	0.44
Teriparatide	944	Year 1	0.78	0.37	0.65
		Year 2	0.53	0.43	0.43

Abbreviation: PDC, proportion of days covered.

TABLE 3 Proportion of days covered for patients initiating treatment in 2017.

	Number of patients initiated in 2017		Mean PDC	Standard deviation	Proportion adherent (PDC ≥ 0.8)
Bisphosphonates	7607	Year 1	0.69	0.37	0.56
		Year 2	0.48	0.43	0.42
Denosumab	8951	Year 1	0.96	0.33	0.75
		Year 2	0.60	0.45	0.47
Teriparatide	783	Year 1	0.74	0.35	0.67
		Year 2	0.49	0.4	0.41

Abbreviation: PDC, proportion of days covered.

patients who are commencing treatment with teriparatide should be prescribed one of the recommended BVMs for teriparatide.²⁵

In February 2024, the HSE agreed to approve reimbursement of romosozumab (Evenity®) for postmenopausal women with severe osteoporosis who have had a major osteoporotic fracture (within the previous year and who are at imminent risk of another fragility fracture).²⁶ The health technology assessment (HTA) summary for romosozumab in the Irish setting estimates a 5-year cumulative gross and net drug budget impact (including VAT) of €15.2 million and €3.2 million, respectively.²⁷ In the HTA, the Marketing Authorisation Holder also included cost offsets, which assumed a reduction in healthcare resource use secondary to reduced incidence of fragility fractures. Accounting for these cost offsets, the 5-year cumulative net healthcare budget impact for romosozumab (including VAT) was estimated to be €3.8 million.²⁷ Reimbursement is subject to a managed access protocol that imposes specific eligibility criteria for reimbursement, specifically restricting reimbursement to a subpopulation of the product licence.²⁸ Managed access protocols are used by the HSE to actively manage the utilization of medicinal products, with the aim of providing a measure of cost containment and cost certainty regarding reimbursement decisions for new medicines.²⁹

Adherence to medications has been defined as the process by which patients take their medications and has three components: initiation, implementation and discontinuation. This study focused on the first two components of medication adherence, initiation and implementation for up to 2 years.³⁰ Persistence with osteoporosis medications both in Ireland and internationally has been shown to be suboptimal, with a range of studies demonstrating 2-year persistence

rates in the region of 50% or below.^{20,31–34} In an Irish study using prescribing data from GP practices, 2-year persistence was found to be 49.4% for oral bisphosphonates and 53.8% for denosumab.²⁰ In a systematic review investigating the persistence and adherence with parenteral osteoporosis treatments, the adherence rates for teriparatide were examined in 29 studies and were 21–89% at 1 year and 37–68% at 2 years. Denosumab was investigated in 19 studies, with 1- and 2-year dose-persistence rates of 61–100% and 36–99%, respectively.³⁵ A study investigating the economic burden of poor adherence with osteoporosis medications found that poor adherence results in a 50% reduction in the potential benefits observed in clinical trials and a doubling of the cost per quality adjusted life year (QALY) gained.³⁴ When denosumab was evaluated for cost-effectiveness compared with bisphosphonates in the Irish setting by the National Centre for Pharmacoeconomics, denosumab had a high probability of cost-effectiveness and reimbursement was recommended.⁵ It is unknown what assumptions were made regarding treatment adherence in that analysis. There are some limitations of the PDC method when assessing medication adherence. For example, as the PDC method considers the proportion of time a patient has medication available, based on prescription claim reimbursement data, it may overestimate adherence by masking non-persistence. This can occur when a patient stops taking the medication for a period (considered non-persistence) and then resumes refilling prescriptions later. In addition, in this study, the PDC method measures whether the medication claim is reimbursed but not whether it is taken as prescribed. Patients might receive dispensed medications but not take them as prescribed, which again may overestimate the adherence value.³⁶

In this study, up to 70% of patients were initiated on denosumab as first-line treatment in the management of osteoporosis. A recent study of Irish GPs cited the vast majority of GP participants (90%) had prescribed denosumab in the last year, with 43% using it as a first-line therapy. Reasons for use of denosumab as first-line therapy included gastrointestinal upset, poor compliance with oral therapy, convenience, severe osteoporosis and renal impairment.³⁷

A global questionnaire study was conducted by the International Osteoporosis Foundation and National Osteoporosis Foundation to ascertain the impact the COVID-19 pandemic had on osteoporosis management from the perspective of healthcare workers. Healthcare workers reported an increase in telemedicine consultations, delays in dual-energy X-ray absorptiometry (DXA) scanning, interrupted supply of medications and reductions in the use of parenteral medications.³⁸ Furthermore, in addition to the effects of COVID-19 on osteoporosis management, measures used to prevent the spread of COVID-19 may have contributed to bone loss. For example, a decrease in physical activity during quarantine and stay-at-home guidance may have contributed to decreased bone mineral density and increased frailty and fall risk, particularly in older adults.³⁹

Our study was based on claims data for medicines dispensed under the CDS; this excludes medicines administered in the hospital setting. Zolendronic acid is an intravenous bisphosphonate, licensed at a dose of 5 mg once yearly for the treatment of osteoporosis and Paget's disease of the bone in adults.⁴⁰ While individual patient level claims data for medicines administered in the hospital setting are not available, unpublished HSE data suggest that approximately 550 units of zolendronic acid 5 mg/100 mL were dispensed through acute hospital pharmacy departments in 2023. This suggests that there is not widespread use of intravenous bisphosphonates for the treatment of osteoporosis in the hospital setting, and provides assurance that the absence of individual patient-level claims data from this setting is unlikely to bias the outcomes of our study.

Our study was conducted using a large, robust national pharmacy claims database. However, it is important to note some limitations. The PCRS database does not include indication for treatment, and we assumed that those in receipt of medicines for the management of osteoporosis had had an osteoporosis diagnosis. As the data included in this study are based on reimbursement claims, there are negligible missing data, however, we could not ascertain if patients had taken medicines as prescribed. It should also be noted that the expenditure data included in this study are representative of ingredient cost alone and exclude patient care fees and VAT. Therefore, the actual state expenditure associated with medicines for the management and treatment of osteoporosis is higher than quoted here. In Ireland, non-oral medicines (such as denosumab and teriparatide) are liable to VAT at the standard rate, currently 23%. Confidential commercial arrangements with marketing authorisation holders, which further reduce the cost of medicines, are excluded from our analyses.

AUTHOR CONTRIBUTIONS

Conceptualization: All authors. *Methodology:* Amelia Smith, Lisa Kelly, and Claire Gorry. *Formal analysis and investigation:* Amelia Smith.

Writing—original draft preparation: Amelia Smith, Lisa Kelly, and Claire Gorry. *Writing—review and editing:* All authors. *Resources:* Michael Barry. *Supervision:* Michael Barry.

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CONFLICT OF INTEREST STATEMENT

There are no competing interests to declare.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study were provided by the HSE-PCRS. The data are not publicly available due to privacy or ethical restrictions.

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APPENDIX 1: MEDICINES PRICES

Denosumab

Prolia Solution for Injection in a Pre-filled Syringe 60 mg/ml 1 mL.
(Pack size 1)

March 2024: €207.51

March 2023: €208.78

March 2022: €209.49

July 2018: €211.95

August 2016: €215.08

Teriparatide

Forsteo Solution For Injection Pre Filled Pen 20 Mcg/80 Mcl
(Pack size 1)

February 2022: €245.14

July 2018: €366.30

August 2016: €374.18

Movymia Solution For Injection Cartridge 20 Mcg/80 Mcl (Pack
size 1)

March 2022: €214.49

Movymia Solution For Injection Starter (2.4 Ml. Cartridge + 1
Pen) Starter Pack 20 Mcg/80 Mcl (Pack size 1)

March 2022: €214.49

Terrosa Solution For Injection Starter (2.4 Ml Cartridge + 1 Pen)
Starter Kit 20 Mcg/80 Mcl (Pack size 1)

March 2022: €214.49

Terrosa Solution For Injection Cartridge 20 Mcg/80 Mcl (Pack
size 1)

March 2022: €214.49

Alendronate 70 mg t

February 2022: €4.32 per 4 tablets

May 2015: €5.04 per 4 tablets

Alendronate 70 mg + colecalciferol 2800 IU

August 2016: €10.65 per 4 tablets

Alendronate 70 mg + colecalciferol 5600 IU

August 2016: €8.09 per 4 tablets

Risedronate 35 mg

January 2015: €5.04 per 4 tablets

Ibandronate 150 mg

October 2024: €5.69 per tablet

Ibandronate Bonviva Tables 150 mg (Pack size 1)

February 2022: €11.98